

Markdown Language Assignment - 01

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Subject: Tools and Techniques for Data Science

Part A — About Me

My name is **Rimsha Mukhtiar**, and I am currently studying **MS Data Science** in my *1st semester* at **Punjab University, Lahore**. I am from **Lahore**, which is also my hometown and the city where I am continuing my studies. My main interests are *programming, reading, and outing*. I enjoy learning new things about technology and improving my programming skills. I am learning Data Science because I am interested in understanding how data can be used to solve real-world problems and help people make better decisions. During my studies, I am exploring different technologies and learning how to work with data using tools such as Python , Pandas , and NumPy . I believe Data Science is an interesting field because it combines programming, mathematics, and problem-solving. My future career goal is to become a **Data Scientist** and develop strong skills in data analysis, machine learning, and data visualization. I hope my studies will help me build useful projects and prepare me for a successful career in the field of Data Science.

Part B — My Technology Skills

Technology	My Level	Experience	I Use It For
Python	Intermediate	Used for small learning tasks and practice programs	Programming, data processing, and basic analysis
NumPy	Intermediate	Used for practice and small data tasks	Numerical calculations and working with arrays

Technology	My Level	Experience	I Use It For
Pandas	Basic	Used for small data analysis tasks	Data cleaning, manipulation, and analysis
Excel	Intermediate	Used for practice and small data tasks	Organizing data, calculations, and basic analysis
SQL	Intermediate	Used while learning database concepts	Writing queries and retrieving data
Power BI	Basic	Used for small practice tasks	Creating dashboards and data visualizations
Matplotlib	Basic	Used for visualization tasks	Creating graphs and charts
Seaborn	Basic	Used for small visualization tasks	Statistical data visualization and exploring datasets

Part C — My Learning Journey

1. Started learning programming basics.
2. Learned Python fundamentals.
3. Created my first simple Python programs.
4. Started working with NumPy.
5. Learned Pandas for data handling.
6. Started learning SQL.
7. Practiced data tasks using Excel.
8. Learned the basics of Power BI.

My Learning Areas

- **Programming**
 - Python
 - Programming Basics
- **Data Handling**
 - NumPy
 - Pandas

- Excel
- SQL
- **Data Visualization**
 - Power BI
 - Matplotlib
 - Seaborn

Part D — My Favorite Technologies

1. Python

- **Why I like it:** Python is easy to understand and useful for Data Science.
- **Where I have used it:** I have used Python for small programming and learning tasks.
- **Online Resource:** [W3Schools Python Tutorial](#)

2. SQL

- **Why I like it:** SQL is useful for working with databases and retrieving data.
- **Where I have used it:** I have practiced SQL queries during my learning.
- **Online Resource:** [SQL Tutorial](#)

3. Excel

- **Why I like it:** Excel is simple and useful for organizing and analyzing data.
- **Where I have used it:** I have used Excel for small data-related tasks.
- **Online Resource:** [Excel Tutorial](#)

4. NumPy

- **Why I like it:** NumPy makes numerical calculations and data operations easier.
- **Where I have used it:** I have practiced NumPy during my learning.
- **Online Resource:** [NumPy Tutorial](#)

5. Pandas

- **Why I like it:** Pandas is useful for handling and analyzing datasets.
- **Where I have used it:** I have used Pandas for small data analysis tasks while learning.

- Online Resource: [Pandas Tutorial](#)

Part E — Project Checklist

Planned Project

Project Topic: Degree–Job Mismatch Analysis

I have recently reviewed the project idea and explored the available data. I am planning to select this topic for my course project. The actual project work has not started yet.

Project Progress

- ☒ Reviewed and explored the project topic
- ☒ Reviewed the available dataset
- ☒ Selected project topic
- ☐ Downloaded dataset
- ☐ Cleaned dataset
- ☐ Started analysis
- ☐ Created visualizations
- ☐ Started report
- ☐ Completed project

Part F — Mathematical Expressions

Mean

$$\text{Mean} = \Sigma x / n$$

Accuracy

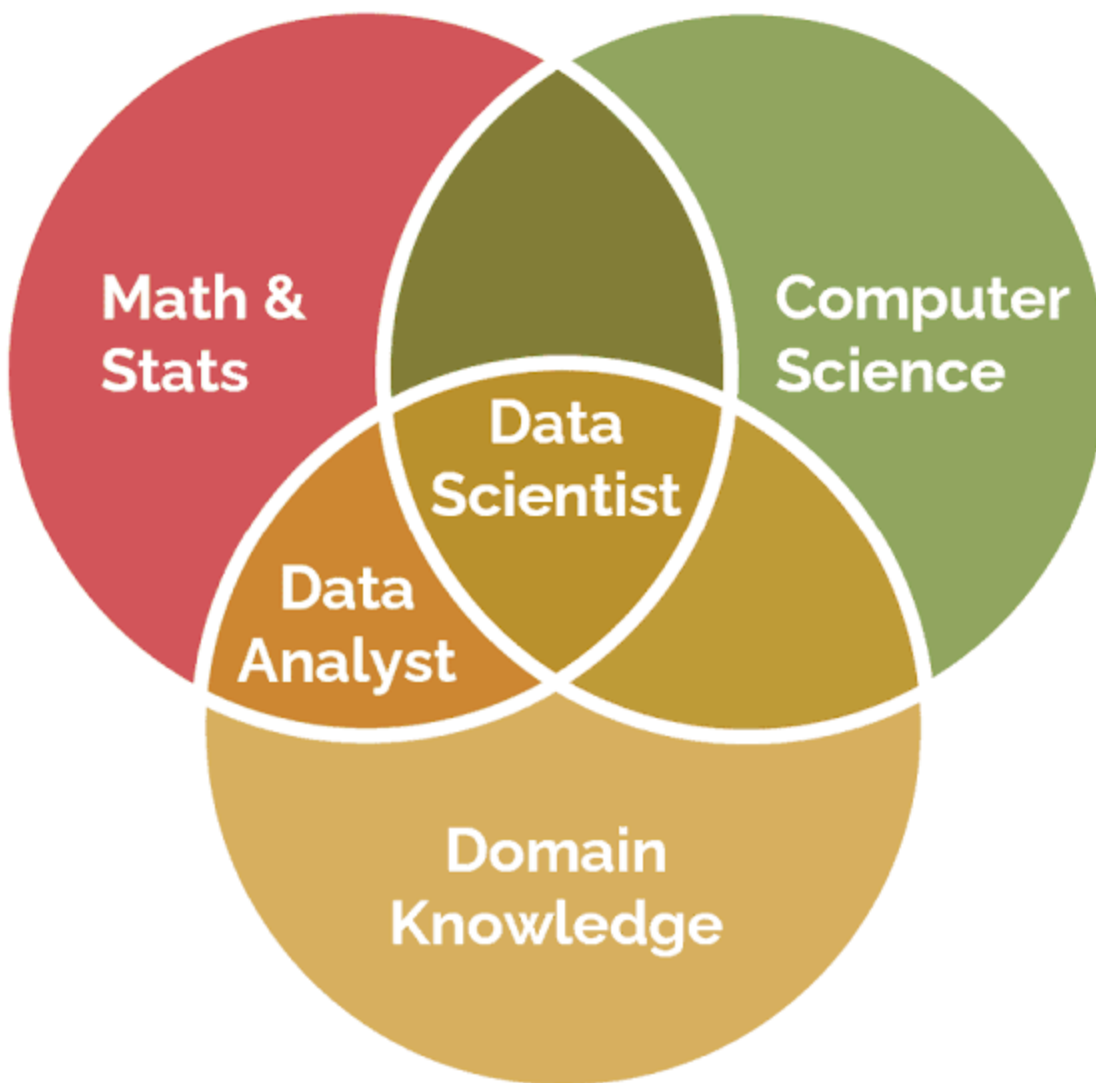
$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Part G — Blockquotes

"The beautiful thing about learning is that nobody can take it away from you." — B.B. King

"Programs must be written for people to read, and only incidentally for machines to execute." —
Harold Abelson

Part H — Data Science Image



Part I — Markdown Features I Practiced

- ☒ Headings
- ☒ Bold
- ☒ Italic
- ☒ Inline Code
- ☒ Ordered List
- ☒ Unordered List
- ☒ Nested List
- ☒ Links
- ☒ Images
- ☒ Tables
- ☒ Blockquotes
- ☒ Mathematical Expressions
- ☒ Checklist
- ☐ Strikethrough
- ☐ Horizontal Rule